THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(a) $\frac{4}{5}$	Hypotenuse is 5
2	(c) $\frac{12}{5}$	Find the perpendicular
3	(b) 1	Substitute the standard values
4	(d) 1	$\tan(90^\circ - \theta) = \cot\theta$
5	(a) 1	$\cos(90^\circ - \theta) = \sin\theta$
6	(d) 4	$k^2(\sin^2 + \cos^2)$
7	(a) $\frac{3}{5}$	Reciprocal
8	(d) $\frac{12}{13}$	Hypotenuse is 13
9	(b) $\frac{15}{8}$	Find the perpendicular
10	(c) $\frac{8}{3}$	Substitute the standard values

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hypotenuse $= 5$ (1); $\sec A = \frac{5}{3}$ (1).	1+1
12	$\cos 65^\circ = \sin 25^\circ$ and $\cot 50^\circ = \tan 40^\circ$ (2); value $= 1 + 1 = 2$ (1).	2+1
13	$\cot 69^\circ = \tan 21^\circ$ (1); value $= 1$ (1).	1+1
14	Hypotenuse $= 13$ (1); $\sec A = \frac{13}{5}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hypotenuse $= \sqrt{12^2 + 5^2} = 13$ (1); $\sin\theta = \frac{12}{13}$, $\cos\theta = \frac{5}{13}$ (1); $\sec\theta = \frac{13}{5}$ (1).	1+1+1
16	$AC = \sqrt{7^2 + 24^2} = 25$ cm (1); $\sin A = \frac{BC}{AC} = \frac{24}{25}$, $\cos A = \frac{AB}{AC} = \frac{7}{25}$ (1); $\tan A = \frac{BC}{AB} = \frac{24}{7}$ (1).	1+1+1
17	$(1 - \cos^2\theta) = \sin^2\theta$ (1); $\sin^2\theta \cdot \csc^2\theta$ (1) $= 1$ (1).	1+1+1
	OR: Hypotenuse $= \sqrt{15^2 + 8^2} = 17$ (1); $\sin\theta = \frac{15}{17}$, $\cos\theta = \frac{8}{17}$ (1); $\sec\theta = \frac{17}{8}$ (1).	1+1+1
18	$AC = \sqrt{20^2 + 21^2} = 29$ cm (1); $\sin A = \frac{BC}{AC} = \frac{21}{29}$, $\cos A = \frac{AB}{AC} = \frac{20}{29}$ (1); $\tan A = \frac{BC}{AB} = \frac{21}{20}$ (1).	1+1+1
	OR: LHS $= \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}$ (2); $= \frac{1}{\sin\theta\cos\theta} = \sec\theta\csc\theta$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	LHS $= \frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$ (1); $= \frac{\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta}{\sin\theta(1+\cos\theta)} = \frac{2 + 2\cos\theta}{\sin\theta(1+\cos\theta)}$ (2); $= \frac{2}{\sin\theta} = 2\csc\theta$ = RHS (2).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{4\tan A + 3}{4\tan A + 3}$ (1); $= \frac{4(\frac{4}{3}) + 3}{4(\frac{4}{3}) + 3} = 1$ (2); verified with $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$ (1).	1+1+2+1
20	LHS $= \left(\frac{1 - \cos\theta}{\sin\theta}\right)^2$ (2); $= \frac{(1-\cos\theta)^2}{1 - \cos^2\theta}$ (1); $= \frac{(1-\cos\theta)^2}{(1-\cos\theta)(1+\cos\theta)} = \frac{1-\cos\theta}{1+\cos\theta}$ (2).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{5\tan A + 2}{3\tan A + 2}$ (1); $= \frac{5(\frac{12}{5}) + 2}{3(\frac{12}{5}) + 2} = \frac{35}{23}$ (2); verified with $\sin A = \frac{12}{13}$, $\cos A = \frac{5}{13}$ (1).	1+1+2+1

Mistakes to avoid:

- Treating $\sin A$ as $\sin \times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta = \frac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.
- Writing $\sin 2A = 2\sin A$ or $\sin(A + B) = \sin A + \sin B$, which are false.

- Wrong table values, e.g. $\sin 30^\circ = \dfrac{\sqrt{3}}{2}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \operatorname{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1	$\tan(90^\circ - \theta) = \cot\theta$
2	**(b)** 1	$\cos(90^\circ - \theta) = \sin\theta$
3	**(d)** 9	$k^2(\sin^2 + \cos^2)$
4	**(b)** $\frac{5}{13}$	Reciprocal
5	**(d)** $\frac{15}{17}$	Hypotenuse is 17
6	**(c)** $\frac{24}{7}$	Find the perpendicular
7	**(c)** 1	Substitute the standard values
8	**(b)** 1	$\tan(90^\circ - \theta) = \cot\theta$
9	**(c)** 1	$\cos(90^\circ - \theta) = \sin\theta$
10	**(b)** 16	$k^2(\sin^2 + \cos^2)$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\cos 72^\circ = \sin 18^\circ$ and $\cot 55^\circ = \tan 35^\circ$ (2); value $= 1 + 1 = 2$ (1).	2+1
12	$\cot 57^\circ = \tan 33^\circ$ (1); value $= 1$ (1).	1+1
13	Hypotenuse $= 17$ (1); $\sec A = \frac{17}{8}$ (1).	1+1
14	$\cos 58^\circ = \sin 32^\circ$ and $\cot 40^\circ = \tan 50^\circ$ (2); value $= 1 + 1 = 2$ (1).	2+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hypotenuse $= \sqrt{24^2 + 7^2} = 25$ (1); $\sin\theta = \frac{24}{25}$, $\cos\theta = \frac{7}{25}$ (1); $\sec\theta = \frac{25}{7}$ (1).	1+1+1
16	$AC = \sqrt{9^2 + 40^2} = 41$ cm (1); $\sin A = \frac{BC}{AC} = \frac{40}{41}$, $\cos A = \frac{AB}{AC} = \frac{9}{41}$ (1); $\tan A = \frac{BC}{AB} = \frac{40}{9}$ (1).	1+1+1
17	Expand: $2\sin^2\theta + 2\cos^2\theta$ (2); $= 2$ (1).	2+1
	OR: Hypotenuse $= \sqrt{21^2 + 20^2} = 29$ (1); $\sin\theta = \frac{21}{29}$, $\cos\theta = \frac{20}{29}$ (1); $\sec\theta = \frac{29}{20}$ (1).	1+1+1
18	$AC = \sqrt{6^2 + 8^2} = 10$ cm (1); $\sin A = \frac{BC}{AC} = \frac{4}{5}$, $\cos A = \frac{AB}{AC} = \frac{3}{5}$ (1); $\tan A = \frac{BC}{AB} = \frac{4}{3}$ (1).	1+1+1
	OR: LHS $= \frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta} = \frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta\cos^2\theta}$ (2); $= \sec^2\theta\csc^2\theta$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Write in $\sin, \cos$: LHS $= \frac{\sin^2\theta}{\cos\theta(\sin\theta - \cos\theta)} + \frac{\cos^2\theta}{\sin\theta(\cos\theta - \sin\theta)}$ (2); $= \frac{\sin^3\theta - \cos^3\theta}{\sin\theta\cos\theta(\sin\theta - \cos\theta)}$ (1); $= \frac{\sin^2\theta + \sin\theta\cos\theta + \cos^2\theta}{\sin\theta\cos\theta} = 1 + \sec\theta\csc\theta$ (2).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{2\tan A + 3}{5\tan A + 1}$ (1); $= \frac{2(\frac{15}{8}) + 3}{5(\frac{15}{8}) + 1} = \frac{54}{83}$ (2); verified with $\sin A = \frac{15}{17}$, $\cos A = \frac{8}{17}$ (1).	1+1+2+1
20	Expand: $\sin^2\theta + 2 + \csc^2\theta + \cos^2\theta + 2 + \sec^2\theta$ (2); $= 5 + \csc^2\theta + \sec^2\theta$ (1); $= 5 + (1 + \cot^2\theta) + (1 + \tan^2\theta) = 7 + \tan^2\theta + \cot^2\theta$ (2).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{3\tan A + 1}{2\tan A + 1}$ (1); $= \frac{3(\frac{24}{7}) + 1}{2(\frac{24}{7}) + 1} = \frac{79}{55}$ (2); verified with $\sin A = \frac{24}{25}$, $\cos A = \frac{7}{25}$ (1).	1+1+2+1

Mistakes to avoid:

- Treating $\sin A$ as $\sin \times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.

- Writing $\tan\theta = \dfrac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.
- Writing $\sin 2A = 2\sin A$ or $\sin(A+B) = \sin A + \sin B$, which are false.
- Wrong table values, e.g. $\sin 30^\circ = \dfrac{\sqrt{3}}{2}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \operatorname{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(a) $\frac{8}{17}$	Reciprocal
2	(a) $\frac{24}{25}$	Hypotenuse is 25
3	(d) $\frac{21}{20}$	Find the perpendicular
4	(a) $\frac{9}{4}$	Substitute the standard values
5	(c) 1	$\tan(90^\circ - \theta) = \cot\theta$
6	(b) 1	$\cos(90^\circ - \theta) = \sin\theta$
7	(b) 25	$k^2(\sin^2 + \cos^2)$
8	(d) $\frac{21}{29}$	Hypotenuse is 29
9	(b) $\frac{40}{9}$	Find the perpendicular
10	(b) 1	Substitute the standard values

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\cot 43^\circ = \tan 47^\circ$ (1); value $= 1$ (1).	1+1
12	Hypotenuse $= 25$ (1); $\sec A = \frac{25}{7}$ (1).	1+1
13	$\cos 76^\circ = \sin 14^\circ$ and $\cot 29^\circ = \tan 61^\circ$ (2); value $= 1 + 1 = 2$ (1).	2+1
14	$\cot 32^\circ = \tan 58^\circ$ (1); value $= 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hypotenuse $= \sqrt{40^2 + 9^2} = 41$ (1); $\sin\theta = \frac{40}{41}$, $\cos\theta = \frac{9}{41}$ (1); $\sec\theta = \frac{41}{9}$ (1).	1+1+1
16	$AC = \sqrt{9^2 + 12^2} = 15$ cm (1); $\sin A = \frac{BC}{AC} = \frac{4}{5}$, $\cos A = \frac{AB}{AC} = \frac{3}{5}$ (1); $\tan A = \frac{BC}{AB} = \frac{4}{3}$ (1).	1+1+1
17	$1 + \tan^2 A = \sec^2 A$ (1); LHS $= (1 - \tan^2 A)\cos^2 A$ (1); $= \cos^2 A - \sin^2 A$ (1).	1+1+1
	OR: Hypotenuse $= \sqrt{8^2 + 6^2} = 10$ (1); $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$ (1); $\sec\theta = \frac{5}{3}$ (1).	1+1+1
18	$AC = \sqrt{12^2 + 16^2} = 20$ cm (1); $\sin A = \frac{BC}{AC} = \frac{4}{5}$, $\cos A = \frac{AB}{AC} = \frac{3}{5}$ (1); $\tan A = \frac{BC}{AB} = \frac{4}{3}$ (1).	1+1+1
	OR: $(1 + \tan^2 A) = \sec^2 A$ (1); $\sec^2 A \sin^2 A = \dfrac{\sin^2 A}{\cos^2 A}$ (1) $= \tan^2 A$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	In sin, cos: $\left(\dfrac{\sin A + \cos A - 1}{\sin A}\right)\left(\dfrac{\cos A + \sin A + 1}{\cos A}\right)$ (2); numerator $= (\sin A + \cos A)^2 - 1 = 2\sin A\cos A$ (2); so value $= 2$ (1).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\dfrac{5\tan A + 4}{5\tan A + 2}$ (1); $= \dfrac{5(\frac{21}{20}) + 4}{5(\frac{21}{20}) + 2} = \frac{37}{29}$ (2); verified with $\sin A = \frac{21}{29}$, $\cos A = \frac{20}{29}$ (1).	1+1+2+1
20	$\dfrac{1}{\cos A}(1 - \sin A) \cdot \dfrac{1 + \sin A}{\cos A}$ (2); $= \dfrac{1 - \sin^2 A}{\cos^2 A}$ (2); $= 1$ (1).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\dfrac{7\tan A + 2}{4\tan A + 1}$ (1); $= \dfrac{7(\frac{40}{9}) + 2}{4(\frac{40}{9}) + 1} = \frac{298}{169}$ (2); verified with $\sin A = \frac{40}{41}$, $\cos A = \frac{9}{41}$ (1).	1+1+2+1

Mistakes to avoid:

- Treating $\sin A$ as $\sin \times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
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- Writing $\tan\theta = \dfrac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.
- Writing $\sin 2A = 2\sin A$ or $\sin(A + B) = \sin A + \sin B$, which are false.

- Wrong table values, e.g. $\sin 30^\circ = \frac{\sqrt{3}}{2}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$.
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